The Contribution of Imports to Consumer Prices*

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Abstract

We develop a new methodology for tracing how border price increases may pass through into US consumer prices. Using this methodology, we show that the 2018 US tariffs passed through fully to consumer prices. We build upon a static open economy model with a network structure, markups, and domestic retailers, from which we carefully map each model object to publicly available US data, including input–output tables and information on industry and retailer margins. We calculate import price sensitivities by expenditure category and predict the partial-equilibrium effects of various tariff scenarios. We find that an additional 10 percent tariff on the rest of the world would prompt a 0.78 to 1.34 percentage point increase in inflation, with significant heterogeneity by country.

Keywords: Tariffs, inflation, import prices, indirect imports
JEL Codes: F40, E65, E31

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Introduction

How do border price fluctuations impact consumer inflation? Though it is clear that, in a partial-equilibrium framework, border price increases should pass through to US consumer prices to some degree, little research to date has mapped and measured the effects of price changes of imported goods on the retail prices of consumption goods.

Our paper develops a new methodology for tracing how import border price increases may pass through into US consumer prices. Our methodology draws on two key insights. First, assessing how import prices affect US consumers requires looking beyond direct consumer purchases of imported products, since domestically produced goods include imported components. Second, due to often large producer and retail markups, it is critical to account for how markups respond to cost changes of imported components.

Our methodology follows a static open economy model (similar to that of Baqaee and Rubbo (2023) and Silva (2024)) of domestic firms that use intermediate domestic and foreign goods, along with domestic labor, in a network structure. For the model to better map to the data, we explicitly add markup and domestic retailers that bundle domestic final goods, foreign final goods, and domestic labor. We carefully lay out how each model object maps to official, publicly available US data.

Our methodology allows us to construct import price sensitivity matrices, which map the purchaser price sensitivity of each consumer expenditure category to changes in the foreign price of each underlying commodity. We show that 10 percent of personal consumption expenditure (PCE) was imported in 2023, and we break down this overall share by expenditure category. Using price changes within each expenditure category over time, we calculate the contribution of import price changes to PCE inflation since 2007.

Importantly, we use our methodology to understand the pass-through of tariffs to consumer prices under different markup response assumptions. Using US Census Bureau data, we calculate the share of each commodity that is imported from each country, allowing us to predict the effect, for any given tariff, on each consumer expenditure category. We then use this calculation to simulate the partial-equilibrium effect of tariff shocks on consumer prices in several different scenarios. We find that when margins remain constant in percentage terms—that is, under full pass-through—tariffs can have nearly double the effect relative to when margins remain constant in dollar terms. There are two reasons for this. First, under full pass-through, positive percentage markups mechanically amplify the effect of an import price shock beyond the share of the good's imported value. Second, the network effects of an upstream tariff propagate downstream, with each producer charging a percentage markup on top of increased suppliers' costs. This theoretical insight is similar to that of Baqaee and

Farhi (2019), who state that when markups are positive, double marginalization divorces producers' revenues from producers' shares in consumer costs.

For example, we consider the tariffs imposed by the United States in 2018. We find that the tariffs would have increased PCE inflation by 0.08 percentage point under constant-dollar markups and by 0.16 percentage point under constant-percentage markups. Furthermore, we simulate the partial-equilibrium effect of an additional 10 percent tariff on all countries in the world as well as on various countries individually. We find that the tariff on all countries would prompt a 0.78 to 1.34 percentage point increase in inflation, with significant heterogeneity when we consider individual countries.

We then ask, how well do our predicted shocks correspond to actual price changes? We address this question using the 2018 tariffs. The first round of tariffs impacted washing machines, aluminum, and steel, while tariffs imposed during the July–September period impacted goods imported from China. It has been particularly difficult for economic researchers to estimate the effects of the 2018 tariffs on consumer prices (Cavallo et al., 2021) because the ways that a tariff increase affects the final consumer is complex and difficult to measure. In addition, micro-level studies based on a difference-in-differences methodology are bound to contain negative omitted variable bias. For instance, suppose we conduct a difference-in-differences comparison between the prices of Chinese shoes and the prices of domestically produced shoes around an enactment of tariffs on Chinese shoes, and suppose we compare prices of the two shoe types within specific retail companies. Negative omitted variable bias will arise from this strategy in two ways. First, goods imported by domestic shoe producers use (directly and indirectly) materials that are in turn affected by tariffs, inflating the price of the "control" group. Second, the protectionist effect of tariffs in the shoe market will endogenously increase prices of domestically produced shoes, given the high degree of substitution within the shoe industry. For both these reasons, a cross-industry difference-in-differences approach is more likely to yield correct estimates because we can account for input–output linkages between industries and because cross-industry elasticities are lower than within-industry elasticities. Indeed, when we compare expenditure categories with various exposures around the China tariffs of 2018, we find full pass-through under our constant-markup assumption. When examining the dynamics of the response, we find that prices rose significantly in the second month of tariffs and continued to rise through the beginning of 2019.

As a final validation exercise, we show that our calculated import sensitivities, along with category-level price changes, allow us to build a consumption-weighted import price index. This index outperforms the US Bureau of Labor Statistics (BLS) Import Price Index, excluding fuel and food, in predicting future inflation, especially at shorter-term horizons of one to three months. Our import price index outperforms the BLS Import Price Index because not only can we adjust for the correct markup assumptions, but also , at its core, our measure better isolates the effect of imports on the consumption component of output, rather than total output.

Related Literature This study contributes to a few strands of the literature. First, it builds upon studies of inflation with production networks (Basu, 1995; Baqaee and Farhi, 2019; La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Baqaee and Rubbo, 2023; Minton and Wheaton, 2023; di Giovanni et al., 2023; Silva, 2024). These studies emphasize the importance of the interaction between sectoral price and wage rigidities and production networks in understanding inflation. Our paper carefully considers how price shocks are disseminated along supply chain networks, taking into account various markup practices.

Second, this paper contributes to the pass-through literature. There is a large literature on exchange rate pass-through (Feenstra, 1989; Goldberg and Knetter, 1997; Amiti, Itskhoki and Konings, 2014), supported by a theoretical literature that explains variable markups through real rigidities (Kimball, 1995; Atkeson and Burstein, 2008). Our research relates most closely to studies examining pass-through in the context of the 2018 US–China trade war. Several studies find that the full burden of the 2018 tariffs was paid by US importers. Using US customs data, Fajgelbaum et al. (2019), Amiti, Redding and Weinstein (2019), and Cavallo et al. (2021) all show full pass-through at the border. However, the literature makes less clear how these higher border prices impacted US consumer prices. Cavallo et al. (2021) uses microdata from two large household retailers to show that these specific retailers had low pass-through of border prices into consumer prices, yet it remains uncertain how consumer prices responded elsewhere. After carefully constructing exposure measures at the consumer expenditure category level (a procedure we first explained in Barbiero and Stein, 2025), we find full pass-through of tariff exposure into consumer prices.[1]

Our study also contributes to the literature on inflation forecasting, as reviewed in Stock and Watson (2009) and Faust and Wright (2013). We supplement the prototypical direct

[1] Since the release of Barbiero and Stein (2025), two policy briefs have been published independently of the present paper. Specifically, Baslandze et al. (2025) and Minton and Somale (2025) focus on empirical estimates of tariff pass-through into consumer prices. Baslandze et al. (2025) use micro-level data linking imports to consumer expenditures. They focus on retail categories available in their data set. We use public data at the industry level to focus on the full effect on the US economy. Moreover, we consider the full network effects of tariffs, not just effects related to directly affected suppliers. Minton and Somale (2025) investigate the pass-through of tariffs into PCE prices. However, they do not consider the importance of markup response assumptions; that is, they compute only the "constant-dollar" percentage markup effect of Barbiero and Stein (2025). Additionally, they do not include all consumer expenditure categories. To our knowledge, our study is the first to evaluate and test the full network effects on PCE inflation under full pass-through. We also include a set of consumer expenditure categories that is broader than those used by existing studies to evaluate the full effect on the US economy.

forecasting model of inflation by including the BLS Import Price Index, excluding fuel and food, and our consumption-weighted import price index as additional regressors. We find that including our consumption-weighted import price index improves a model that uses only past inflation as well as a model that also uses the official import price index.

The paper is organized as follows. Section 1 presents our partial-equilibrium network model. Section 2 describes the data we use, all of which are publicly available, and how we map the model to the data to construct the import price sensitivity of expenditure categories. Section 3 discusses a few applications of our methodology: First, it describes the import share of consumption and breaks down this share by category; second, it uses our import price sensitivities to simulate consumer price responses to tariff shocks; third, it compares these predicted inflation outcomes with category-level inflation around the 2018 tariffs. Section 4 presents a validation exercise for our methodology in which we forecast inflation with the consumption-weighted import price index. Section 5 concludes.

1 Model

Our methodology allows us to map—dollar for dollar—the impact of a border price increase to the most disaggregated expenditure categories available in the US consumption data. To help interpret our methodology and its assumptions, we build a static open economy model in partial equilibrium, with two primary factors: import goods and labor. In the domestic economy, there are two kinds of producers: retailers and firms. Firms use intermediate domestic and foreign goods, as well as labor, to produce domestic goods. Retailers bundle domestic final goods and foreign final goods with domestic labor. Households consume composite retail goods. Due to market structures, domestic goods producers and retailers both incorporate a markup into their price. Without loss of generality, trade is balanced. The model is similar to that of Baqaee and Rubbo (2023) and Silva (2024), except we explicitly add markup and retailers to improve mapping with the data.

Households We assume households have homothetic utility of various consumption goods bundled by retailers and indexed by e. This utility, written as $U = U(C_1, ..., C_e, ..., C_E)$, is subject to the following budget constraint

$$\sum_{e=1}^{E} p_e c_e = wL + \sum_{e=1}^{E} \pi_e + \sum_{i=1}^{N} +\pi_i$$

In other words, households earn wages w for their labor L, and retail and domestic-goods producers' profits are rebated to households.

Retailers The producers of C_e are monopolistically competitive retailers that combine domestic final goods indexed by i and foreign final goods indexed by i^* with the following Hicks-neutral constant returns to scale production function:

$$x_e = A_e F^e(L_e, \{x_{ei}\}_{i=1,\ldots,N}, \{x_{ei^*}\}_{i^*=1,\ldots,N^*}),$$

where A_e is productivity, L_e is labor, and x_{ei} and x_{ei^*} represent final goods quantities used in production.

The cost minimization problem of retailers is

$$\min_{L_e,\{x_{ef}\}_{f=1,\ldots,N},\{x_{ei^*}\}_{i^*=1,\ldots,N^*}} w_e L_e + \sum_{i=1}^{N} p_i x_{ei} + \sum_{i^*=1}^{N} p_{i^*} x_{ei^*}, \tag{1}$$

such that $x_e \geq \bar{x_e}$. Retailers do not use other retailers' goods to generate their aggregate bundle x_e. This means that there is no network structure to track *within* the retailer market. Each producer e's optimal price in equilibrium is

$$p_e = (1+\mu_e)MC_e = (1+\mu_e)\frac{TC_e}{x_e} = \frac{(1+\mu_e)}{A_e}\left[\frac{w_e L_e}{x_e} + \sum_{i=1}^{N}\frac{p_i x_{ei}}{x_e} + \sum_{i^*=1}^{N^*}\frac{p_{i^*} x_{ei^*}}{x_e}\right],$$

where μ_e denotes markup, MC_e denotes marginal cost, TC_e denotes total cost, w_e denotes wages, and p_i and p_{i^*} are the prices of x_{ei} and x_{ei^*}, respectively.

The preceding expression simplifies to the following first-order expression, where $d\log y = \hat{y}$,

$$\hat{p}_e = \hat{\mu}_e - \hat{A}_e + (1+\mu_e)\theta_e^L \hat{w}_e + \sum_i (1+\mu_e)\theta_{ei}\hat{p}_i + \sum_{i^*}(1+\mu_e)\theta_{ei^*}\hat{p}_{i^*},$$

where θ_e^L is the sales-based share of factor L defined as $\theta_e^L = \frac{w_e L_e}{p_e x_e}$, $\theta_{ei} = \frac{p_i x_{ei}}{p_e x_e}$ is the sales-based share of domestically produced final good i sold by retailer e, and $\theta_{ei^*} = \frac{p_{i^*} x_{ei^*}}{p_e x_e}$ is the sales-based share of imported good i^* sold by retailer e. Since the preceding expression holds for each retailer, we express it in matrix form for all retailers as

$$\hat{P}^E = \hat{M}^E - \hat{A}^E + M^E\left(\Theta^L \hat{W}^E + \Theta\hat{P} + \Theta^*\hat{P}^*\right), \tag{2}$$

where $\hat{P}^E$ is a $1 \times E$ vector of all retail price changes, $\hat{M}^E$ and $\hat{A}^E$ are shocks to markups and productivity of retailers, and M^E is an $E \times E$ diagonal matrix of retail markups. The Θ matrices are $E \times N$ sales-based factor shares matrices. $\hat{W}^E$, $\hat{P}$, and $\hat{P}^*$ are wage and producer-price changes. Note that capital letters are either vectors or matrices. If we keep constant all exogenous primitive factor prices and productivity, except final goods prices, the preceding expression collapses to

$$\hat{P}^E = M^E \left(\Theta \hat{P} + \Theta^* \hat{P}^* \right). \tag{3}$$

Domestic Goods The monopolistically competitive producers of the domestic good i's cost minimization function is

$$\min_{L_i, \{x_{ij}\}_{j=1,\dots,N}, \{x_{ij^*}\}_{j^*=1,\dots,N}} w_i L_i + \sum_{j=1}^{N} p_j x_{ij} + \sum_{j^*=1}^{N^*} p_{j^*} x_{ij^*}, \tag{4}$$

such that $x_i = A_i F^i(L_i, \{x_{ij}\}_{j=1,\dots,N}, \{x_{ij^*}\}_{j^*=1,\dots,N}) \geq \bar{x}_i$. Thus, unlike with retailers, there is a network structure within domestic goods production.

Each producer i's optimal price in equilibrium is

$$p_i = (1+\mu_i) MC_i = (1+\mu_i) \frac{TC_i}{x_i} = \frac{(1+\mu_i)}{A_i} \left[\frac{w_i L_i}{x_i} + \sum_{j=1}^{N} \frac{p_j x_{ij}}{x_i} + \sum_{j^*=1}^{N^*} \frac{p_{j^*} x_{ij^*}}{x_i} \right],$$

This approximate to the following first-order equation

$$\hat{p}_i = \hat{\mu}_i - \hat{A}_i + (1+\mu_i) \frac{w_i L_i}{p_i x_i} \hat{w}_i + \sum_j (1+\mu_i) \frac{p_j x_{ij}}{p_i x_i} \hat{p}_j + \sum_{j^*} (1+\mu_i) \frac{p_{j^*} x_{ij^*}}{p_i x_i} \hat{p}_{j^*}.$$

We can express this equation as a function of sales-based factor shares for a factor r, $\Omega^r_{ij} = \frac{p^r_j x^r_{ij}}{p_i x_i}$:[2]

$$\hat{p}_i = \hat{\mu}_i - \hat{A}_i + (1+\mu_i) \Omega^L_i \hat{w}_i + \sum_j (1+\mu_i) \Omega_{ij} \hat{p}_j + \sum_{j^*} (1+\mu_i) \Omega^*_{ij} \hat{p}_{j^*}.$$

Since the equation holds for every product i, we express it in matrix form:

$$\hat{P} = \hat{M} - \hat{A} + \Omega^L \hat{W} + M \Omega \hat{P} + M \Omega^* \hat{P}^*,$$

where $\hat{P}$ is the $N \times 1$ vector of domestic price changes, and $\hat{M}$ is the $N \times 1$ vector of markup shocks. Ω^L is the sales-based vector of labor costs, which is a diagonal matrix in this model, though this assumption can be relaxed. Ω is the matrix of intermediate goods cost, while

[2]Alternatively, we can express the equation in terms of cost-based factor shares $\tilde{\Omega}^r_{ij} = (1+\mu_i)\Omega^r_{ij} = \frac{p_i}{MC_i} \frac{p^r_j x^r_{ij}}{p_i x_i} = \frac{p^r_j x^r_{ij}}{TC_i}$:

$$\hat{p}_i = \hat{\mu}_i - \hat{A}_i + \tilde{\Omega}^L_i \hat{w}_i + \sum_j \tilde{\Omega}_{ij} \hat{p}_j + \sum_{j^*} \tilde{\Omega}^*_{ij} \hat{p}_{j^*}$$

Ω^* is the import factor share matrix. M is a diagonal matrix, where each element of the diagonal is the good-specific markup.

Inverting the system gives

$$\hat{P} = (\mathbb{I} - M\Omega)^{-1} \left(\hat{M} - \hat{A} + \Omega^L \hat{L} + M\Omega^* \hat{P}^* \right). \tag{5}$$

The preceding equation represents the general case. If we keep shocks to productivity, markups, and wages constant, this collapses to

$$\hat{P} = (\mathbb{I} - M\Omega)^{-1} M\Omega^* \hat{P}^*. \tag{6}$$

Plugging (6) into (3), we obtain

$$\hat{P}^E = M^E(\Theta^* + \Theta(\mathbb{I} - M\Omega)^{-1} M\Omega^*)\hat{P}^*. \tag{7}$$

2 Mapping the Model to the Data

As a contribution in its own right, this study carefully maps each model object to official US public data. This section provides intuition about the available data and the economic concepts behind them.

Our model shows that the impact of a vector of import price changes $\hat{P}^*$ on the vector of retail final goods price changes $\hat{P}^E$ can be decomposed into $\hat{P}^E = (\Delta^{\text{Direct}} + \Delta^{\text{Indirect}})\hat{P}^*$, where

$$\text{Direct Sensitivity:} \quad \Delta^{\text{Direct}} = M^E \Theta^* \tag{8a}$$

$$\text{Indirect Sensitivity:} \quad \Delta^{\text{Indirect}} = M^E \Theta (\mathbb{I} - M\Omega)^{-1} M\Omega^*. \tag{8b}$$

The direct sensitivity is the predicted response to an import price increase of consumer prices due to the share of personal expenditures in foreign *final* goods. The indirect sensitivity is the predicted response of consumer prices due to the share of domestic final goods consumption that uses imported intermediate goods in production processes.

We now unpack the intuition behind each component in equations (8a) and (8b), and we show how to map these objects to the data. We sourced our data from several publicly available tables from the US Bureau of Economic Analysis (BEA) Input–Output Accounts, namely the Use Tables, the Make Tables, the Import Matrices, and the PCE Bridge.[3]

[3]In all cases, we use the "after redefinition" versions of the tables.

- Θ and Θ^* are the domestic and foreign goods retail aggregation matrices, respectively. They are $E \times N$ matrices (where E is the number of retail goods and N is the number of final goods) containing the sales-based shares of domestic or imported products, respectively, purchased by retailers to produce each retail good.

  In practice, Θ and Θ^* must be built by merging a few separate BEA data sets. First, the shares of domestic and imported commodities in final demand are calculated from the Use Tables and Import Matrices. (In this paper, we use the terms "commodity" and "final good" interchangeably—both are the direct inputs into the retail sectors, but "commodity" is the term used in the BEA data.) Specifically, each table includes a column containing final demand, either in total or for imports, of each commodity in producers' prices. Second, these (BEA code) commodities are concorded to each retail-level (national income and product accounts [NIPA] code) final expenditure category using the PCE Bridge.

- Ω and Ω^* are the domestic and import expenditure matrices, respectively. They are $N \times N$ matrices, where each cell Ω^x_{ij} contains the share of domestic or foreign commodity j used to produce the domestic commodity i. In practice, Ω and Ω^* are built by starting with the domestic and foreign direct requirement matrices, respectively, which show the input of each commodity required per unit of gross output of each industry. These matrices are obtained from the commodity-by-industry Domestic Use Table and Import Table and are normalized by industry gross output. Both matrices must then be multiplied by the proportion of total output of each commodity produced in each industry, obtained from the Make Table, converting these matrices into square commodity-by-commodity matrices.[4]

- M^E is the retail markup matrix. It is an $E \times E$ diagonal matrix containing retail-level markups specific to each retail sector. Under full pass-through (described in detail later), M^E is built starting from the PCE Bridge, which gives the value added of each retail sector from each commodity by listing both producer and purchaser prices. Furthermore, this data set decomposes this value added into the contribution of the transportation, wholesale, and retail sectors. We net employee compensation from the value added of each of these three sectors using information from the Use Tables.

- M is the commodity markup matrix. It is an $N \times N$ diagonal matrix containing

[4] Note that all theoretical matrices follow the general convention in the literature to have outputs in the row and inputs in the columns of the factor share matrix Ω. The BEA instead specifies inputs in the rows and outputs in the columns of the use tables. In addition, the BEA distinguishes between commodity and industry in its input–output table.

commodity-level markups specific to each domestic production sector. Under full pass-through, M is computed by dividing gross output by the total variable costs of each commodity, all obtained from the Use Table.[5]

We can now see how the direct sensitivity is thus the share of expenditure of foreign final goods within each retail category, augmented by the size of the markup in each retail category. We can also interpret the indirect sensitivity as the amount of imported cost increases for intermediate purposes (that is, $M\Omega^*$), propagated through the domestic Leontief matrix (that is, $(\mathbb{I} - M\Omega)^{-1}$) and bundled into domestic retail goods (that is, $M^E\Theta$). In practice, these sensitivities are year-specific, but we omit the time subscript for simplicity. We later describe how we calculate sensitivities for each year.

In our sensitivity calculations, we use two main competing pass-through assumptions:

- **Constant-percentage Markup** (or full pass-through): Domestic markups over marginal costs are assumed to be constant in percentage terms, in line with both our model and the standard benchmark in the economic literature. We allow for production- and retail-sector-specific markups by populating the matrices M and M^E as described earlier.

- **Constant-dollar Markup** (or partial pass-through): Domestic profits are assumed to be constant in nominal value after an import price increase. Under this assumption, markup can be considered a domestic factor of production that is exogenous to import price changes. This is equivalent to assuming partial pass-through because profits *decrease* in percentage terms over marginal costs but not enough to fully absorb the impact of additional costs for the consumer. Another appealing feature of this assumption is that the preceding import sensitivity matrices can be interpreted as shares of imported value in final expenditure. In practice, we substitute the markup matrices M and M^E with identity matrices to when considering this case.

Finally, we define the importance-weighted PCE sensitivity matrix as

$$\Delta^w = W^c(\Delta^{\text{Direct}} + \Delta^{\text{Indirect}}), \tag{9}$$

where W^c is a diagonal matrix with diagonal values defined as the share of each PCE category in total PCE. Aggregating the importance-weighted sensitivity by imported commodity, as in $\mathbf{1}^\top\Delta^w$, gives the aggregate sensitivity of personal consumption expenditures to each imported commodity. Aggregating the importance-weighted sensitivity by expenditure category, as in $\Delta^w\mathbf{1}$, gives the importance-weighted sensitivity to total imports of each expenditure category.

[5] Total variable costs are the sum of salaries and cost of materials.

Interpolation of Data from More Detailed Years BEA input–output tables are published annually A coarse version of each table is available for 1997 through 2023 and includes 71 industry groups and 73 commodity groups. Finer versions, with 402 industries and 402 commodities, are available for 2007, 2012, and 2017. Similarly to the input–output account matrices, the Bridge files also come in two versions: a coarser version with annual updates for 1997 through 2023 using 73 commodity groups and 76 NIPA expenditure groups, and a finer version for 2007, 2012, and 2017 using 402 commodities and 212 NIPA expenditure codes. We use the coarser level of aggregation in the yearly data and multiply the coarse tables by the interpolated cell-block share trends from the detailed tables to populate input–output tables at the most disaggregated level for all years from 1997 through 2023.

3 Results

3.1 Import Shares

Our import sensitivities under the constant-dollar markup (partial pass-through) assumption can be reinterpreted as the share of US PCE that is imported.[6] Summing across all PCE categories, we find that 6 percent of core PCE was directly imported in 2023, and 4 percent was indirectly imported. In total, spending on direct and indirect imports accounted for 10 percent of total core PCE.[7] This percentage includes US components of foreign goods that are then imported into the United States.

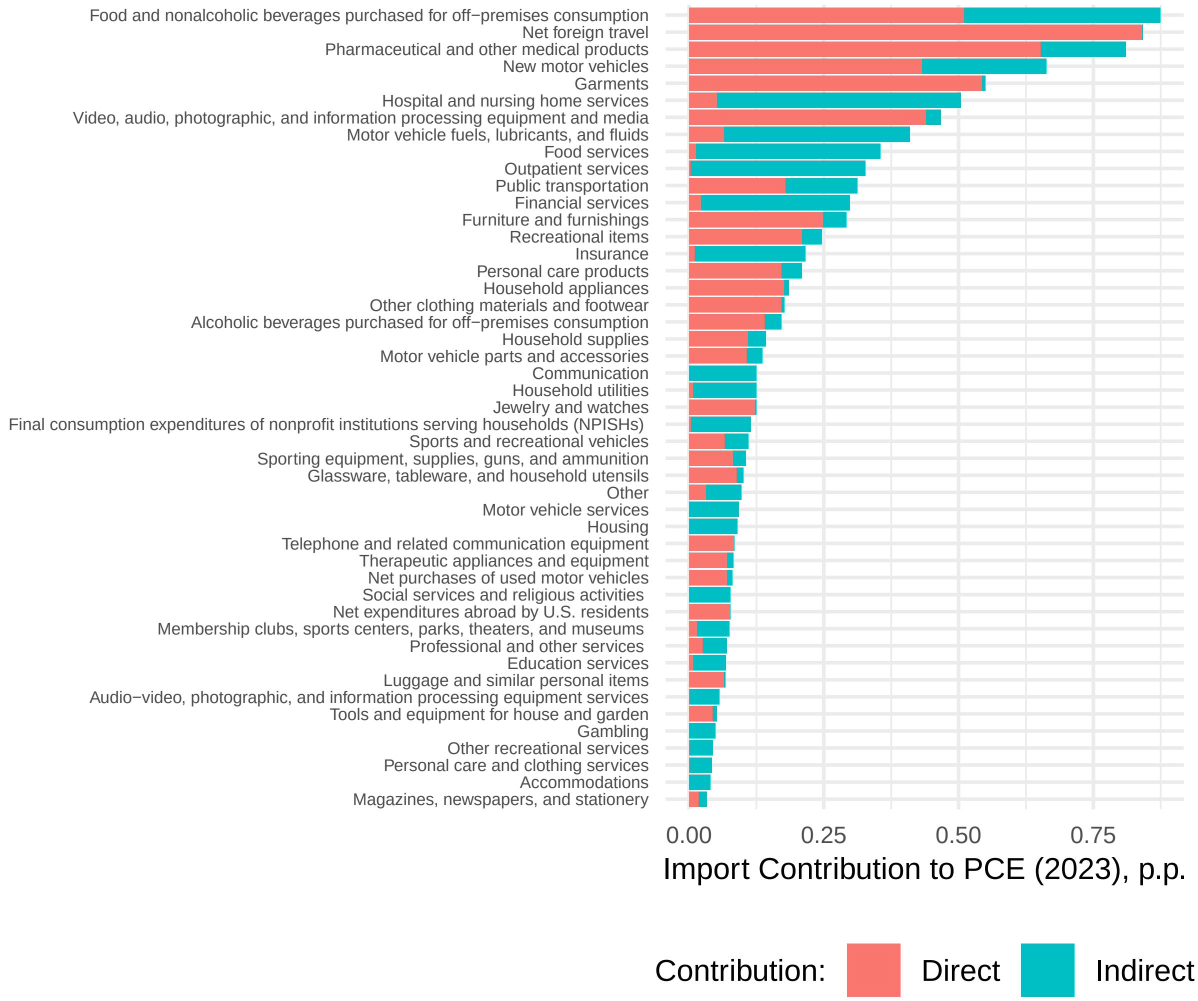
These shares have been consistent over the past two decades.[8] Our methodological framework enables us to break down these shares into underlying input commodities or expenditure categories. Figure 1 shows the contribution of both direct and indirect imports to PCE, broken down by NIPA expenditure category, and weighted by their PCE importance, as in equation 9. For example, *direct* imports of pharmaceutical and other medical products account for 0.65 percent of PCE. Spending on *indirect* imports used in the domestic production of pharmaceutical and other medical products comprises 0.16 percent of PCE. Total imports—the sum of the direct and indirect components—in pharmaceutical and other medical products consumption account for 0.81 percent of PCE. The figure also highlights that some expenditure categories, such as hospital and nursing home services, may use a large quantity of indirect imports, though these services tend to be domestically produced.

[6] Under the constant-dollar markup assumption, markups do not change in dollar terms with import price changes and thus can be thought of as a domestic factor of production.

[7] Our calculation of the import contribution to PCE is similar but not equivalent to that of Hale, Hobijn and Fernanda Nechio (2019), who estimate that 11 percent of headline PCE can be traced to imported goods.

[8] See Appendix Figure A.1.

Figure 1: PCE Shares of Imports, by Expenditure Category



Notes: The red bars show the share of PCE corresponding to imported goods that are directly consumed by US households. The blue bars show the share of PCE corresponding to imports that are used as inputs in domestic production. We use the NIPA categories of goods and services. *Sources*: US Bureau of Economic Analysis and authors' calculations.

3.2 Simulation of Tariff Shocks

In addition to offering a one-to-one mapping between imports and personal consumption expenditures, our methodology comprises a straightforward tool for studying the inflationary effect of specific tariff episodes. In this section, we outline the importance of the markup assumption response in determining the total impact on PCE inflation. Additionally, we show how our methodology is useful in intuitively assessing the impact of tariff exemptions, a feature of most tariff reforms.

To do so, we build a concordance matrix between product codes available from border declarations and BEA commodity codes.[9] We define F as a matrix containing the share of country-product specific value of import for each BEA commodity. Each column represents a country-HTSUS combination, while each row represents one of the 402 BEA commodities. We then use a country-product-specific vector of tariffs τ_t at any given time to predict the aggregate tariff effect as

$$\hat{P} = \mathbf{1}^\top W^c(\Delta^{\text{Direct}} + \Delta^{\text{Indirect}})F\tau. \tag{10}$$

The foreign trade concordance matrix F offers a convenient bridge to simulate the effects of tariffs or border shocks τ starting at the most disaggregated level possible.[10] Such level of disaggregation in the tariff scenario τ allows us to account for exemptions, which are usually set at the 10-digit HTSUS-country level by law. Our methodology allows for an exempted category to have both a direct (null) effect on consumption purchases via Δ^{Direct} and an indirect (null) effect via the domestic supply chain.[11]

We show the effect of different tariff scenarios under our two main markup response assumptions introduced in Section 2: constant-dollar markup for partial pass-through and constant-percentage markup for full pass-through. Our estimates assume that the full burden of the tariffs is paid by US importers, consistent with what was observed with the tariff increases imposed in 2018 (Fajgelbaum et al., 2019; Amiti, Redding and Weinstein, 2019; Cavallo et al., 2021).

Figure 2 shows the PCE inflation effect under three different tariff scenarios. First, we use the cumulated effect of all tariffs imposed on US imports in 2018.[12] We find a predicted partial-equilibrium effect on headline PCE inflation of 0.08 percentage point under constant-dollar markup or 0.16 percentage point under constant-percentage markup. Most of the effect comes via the indirect channel, that is, from higher costs of US domestic producers. This may be why earlier studies struggle to find a pass-through of these tariffs into consumer prices (Cavallo et al., 2021). In the next section, we show how these estimates are empirically confirmed in cross-sectional PCE responses.

Second, we assess a hypothetical uniform 10 percent tariff applied to all imports. We find that this scenario leads to an increase in headline PCE prices of 0.78 percentage point under constant-dollar markup or 1.34 percentage points under constant-percentage markup. Third, we examine a uniform 10 percent tariff on all imports but with a US content exemption.

[9] We use the US Census Bureau Foreign Trade import data, from which we calculate the total dollar value imported into the United States of each HTSUS commodity code from each foreign country in each year.

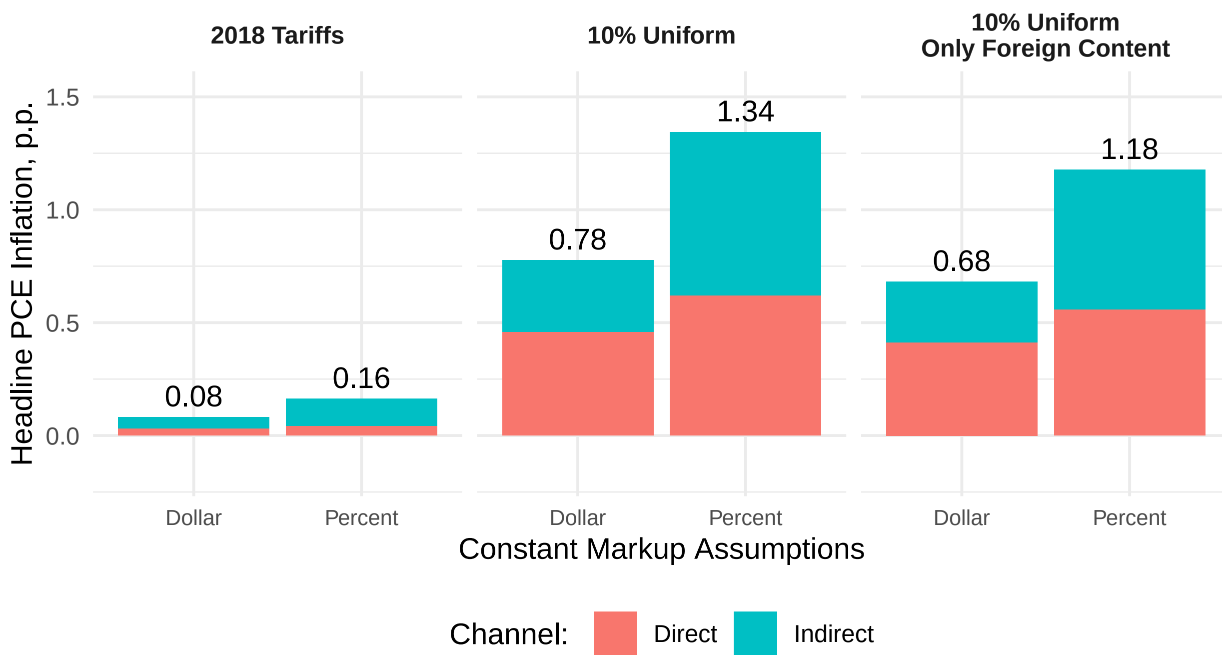
[10] As in the previous sections, all these matrices are year-specific, but we omit the time subscript for clarity.

[11] For example, say that mechanical components are exempted from a certain tariff but steel is not exempted. This methodology would account for the final price increase of mechanical components due to the increased price of steel, to the extent that steel is used to produce mechanical components.

[12] These tariffs, and related exemptions, were provided by Fajgelbaum et al. (2019) at the country-product level.

We find this exemption makes only a marginal difference of 0.1 to 0.2 percentage point.[13] We find an increase in headline PCE prices of 0.68 percentage point under constant-dollar markup or 1.18 percentage points under constant-percentage markup.

Figure 2: Headline PCE Impact under Different Tariff Scenarios



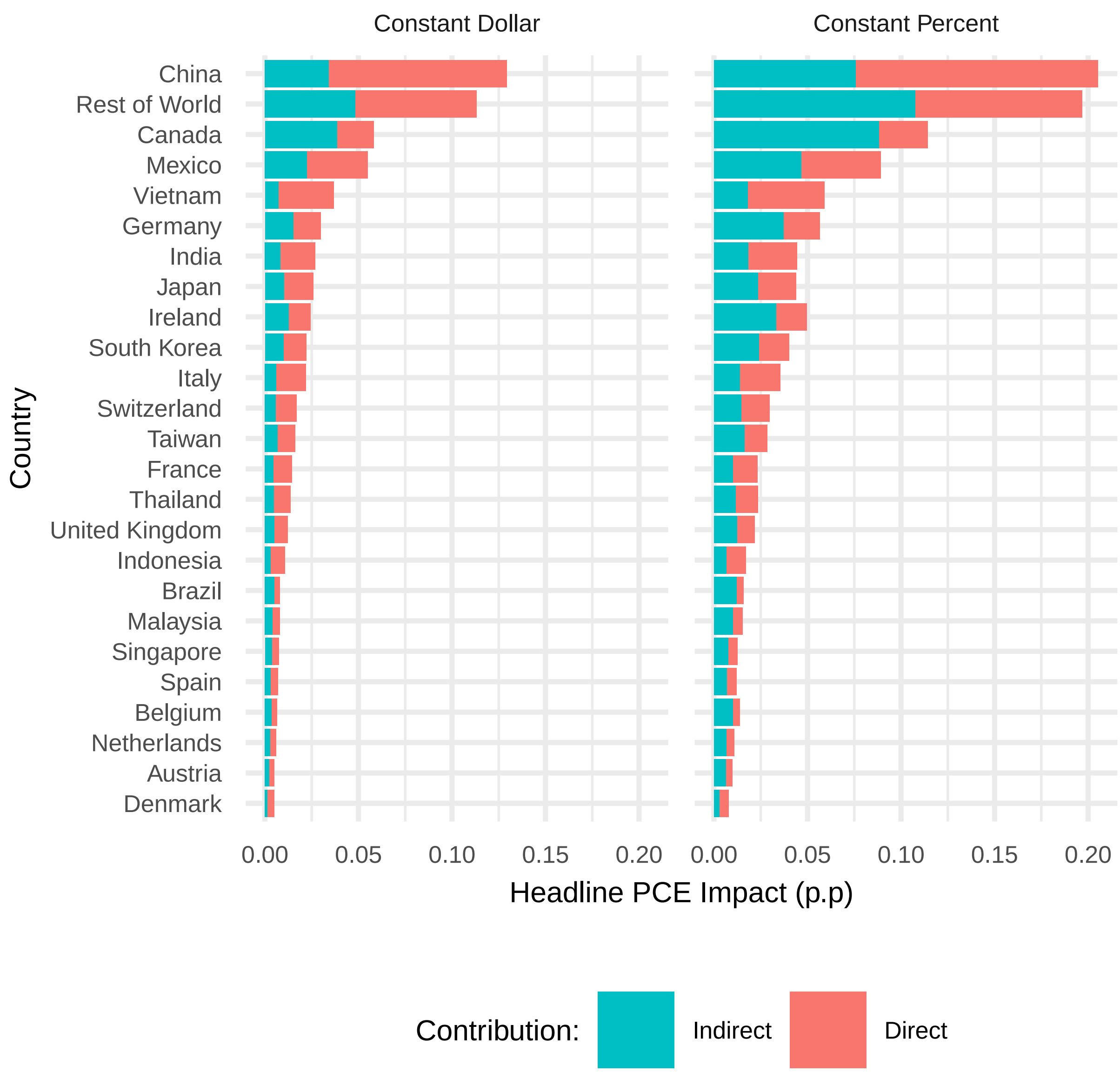
Notes: The three tariff scenarios are computed from equation (10). In the first scenario, we use as τ_1 the cumulated tariffs introduced by the Trump Administration in 2018 and gathered at the country-HTSUS level by Fajgelbaum et al. (2019). In the second scenario, we use $\tau_2 = 10\%$. In the third scenario, we use $\tau_3 = (\mathbf{1} - s^{\mathrm{US}}) \oslash 10\%$, where s^{US} is the US content in US imports computed for each country-ISIC product combination from the OECD 2020 TiVA database. We use the share of USMCA imports to discount the tariff effects for Canada and Mexico instead.
Sources: BEA, US Census Bureau, Fajgelbaum et al. (2019), OECD TiVA, and authors' calculations.

Two notable results are depicted in Figure 2. First, given the importance of markups in the data, assuming constant-percentage markup can nearly double the effect that is predicted when assuming constant-dollar markup. Second, we find that the increased effect under constant-percentage markup acts primarily through indirect effects. The intuition is that an upstream marginal cost increase due to a tariff is amplified along the supply chain, as each producer charges a constant-percentage markup on top of each respective input purchase.

[13] Under this exemption, the tariff is not applied on the percentage of value added originated in the United States. This is a typical exemption included in most executive orders introduced by the Trump Administration in 2025. Our methodology can easily account for such exemptions using the OECD input–output tables. In particular we discount for the US content in each country's export to the United States. These shares can vary widely for each country-product combination. In the case of Canada and Mexico, we use the share of USMCA-compliant goods instead.

This outsized effect of domestic markups on indirect imports is particularly visible when we disaggregate such effects by source country.

Figure 3: Source Country Decomposition of a 10 Percent Uniform Tariff with US-content exemption



Notes: Each line represents the country-specific contribution to the total PCE tariff effect computed, as in equation (10), where $\tau = (\mathbf{1} - s^{\text{US}}) \oslash 10\%$. s^{US} is the US content in US imports computed for each country-ISIC product combination from the OECD 2020 TiVA database. We use the share of USMCA imports to discount the tariff effects for Canada and Mexico instead.
Sources: BEA, US Census Bureau, and authors' calculations.

Figure 3 shows the decomposition of a 10 percent uniform tariff with US-content exemption by goods' origin country. All effects are weighted by the importance of each category within total PCE. China is the top source of the inflationary effect, due mostly to direct purchases from consumers. However, the indirect effect becomes more important under the constant-percentage markup assumption. Canada and Mexico are also important inflationary sources under a 10 percent tariff, despite the tariff discount on USMCA-compliant goods.

Note that our methodology implies that the effect grows linearly to the simulated tariff vector τ (conditional on the markup assumption). For instance, since Figure 3 implies that a 10 percent tariff with US-discounted content on Chinese goods has a 0.12 to 0.21 percentage point effect on PCE inflation, a similar 20 percent tariff will have an effect of 0.24 to 0.42 percentage point.

3.3 Past Tariff Shock Pass-through into Consumer Prices

As discussed in the previous section, our methodology allows us to predict the effect of tariff shocks on PCE inflation in a partial-equilibrium framework. It is natural to then ask, how have our predicted shocks, based on calculated import price sensitivities, actually passed through to PCE during past episodes? The answer may validate our markup assumption.

To address this question, we zoom in on the 2018 tariffs and focus individually on NIPA expenditure groups.[14] We focus on the China tariffs enacted during the July–September 2018 period. We collapse the different waves of tariffs into one tariff shock, and we run a difference-in-differences specification in which we compare year-over-year price changes of expenditure categories with various predicted tariff effects. Formally, we run the following regression model:

$$\pi_{i,t} = \alpha_i + \delta_t + \beta \text{Post-Period}_t \times s_i + \gamma \hat{w}_{i,t-1} + \varepsilon_{i,t}, \tag{11}$$

where π denotes the year-over-year price change,[15] s denotes the predicted effect for NIPA expenditure category i (as explained in subsection 3.2), and $\hat{w}$, which is included as a control variable, denotes the year-over-year wage change.[16] Predicted effects are calculated using the tariff data in Fajgelbaum et al. (2019). i refers to the NIPA expenditure category, and t refers to the monthly date. The post-period begins in July 2018, and our sample runs from July 2017 through April 2019. We choose this end date to avoid contamination with the tariffs imposed on China starting in May 2019. Our regression also includes NIPA and time

[14]The analyses in this section use NIPA expenditure groups at the level shown in Figure 1.

[15]We take year-over-year price changes to control for seasonality.

[16]The wage control comes from the BLS Current Employment Statistics (CES) data, specifically the seasonally adjusted average hourly earnings of all employees series. The data are at the NAICS level, so to get them to the NIPA level, we follow a procedure similar to the one outlined in Section 4.

fixed effects. The key parameter of interest is β, which approximates the pass-through. A coefficient of 1 would indicate full pass-through.

As shown in Table 1, for goods, there seems to have been full pass-through when we use the constant-percentage markup assumption, indicated by the significant coefficient of 0.968, only slightly less than 1. If we use the constant-dollar assumption, the pass-through coefficient is greater than 1. This is likely due to a predicted effect that undershoots, on avaerage, the actual PCE response under constant-dollar markup. This evidence favors our constant-percentage markup assumption, even though we cannot reject the hypothesis that the constant-dollar coefficient is statistically different from 1. Turning to services, we see weaker results, with positive but insignificant coefficients.

Table 1: Pass-through Depending on Markup Assumption

	Goods		Services	
	(1)	(2)	(3)	(4)
Post-period × Predicted Effect	0.968**		0.621	
Assumption: Constant-Percent	(0.434)		(1.237)	
Post-period × Predicted Effect		1.253**		1.252
Assumption: Constant-Dollar		(0.600)		(1.680)
Wage change	-0.001	-0.002	0.004	0.005
	(0.029)	(0.029)	(0.035)	(0.037)
NIPA FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
R^2	0.644	0.643	0.690	0.690
R^2 within	0.024	0.023	0.006	0.007
N	594	594	506	506

Notes: This table shows the pass-through from the China tariffs enacted during the July–September 2018 period. The post-period begins in July 2018, and our sample runs from July 2017 through April 2019. Standard errors are clustered at the NIPA expenditure category level. *Sources*: BEA, BLS, Fajgelbaum et al. (2019), authors' calculations.

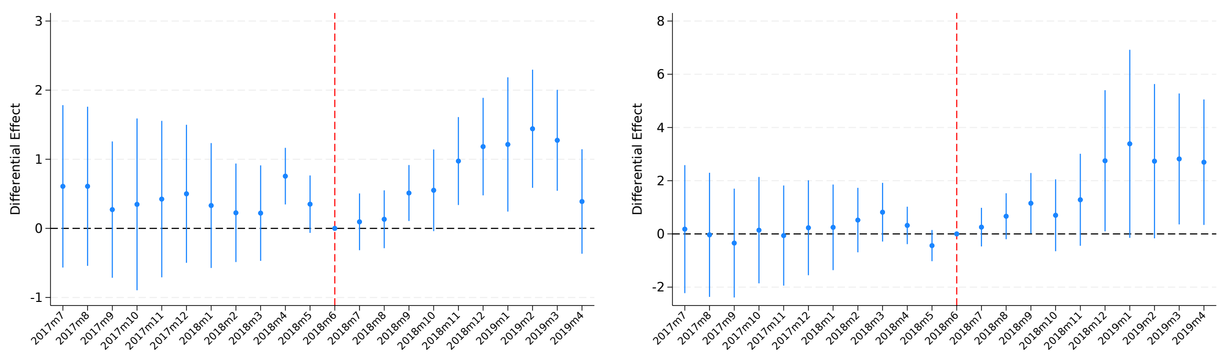
We next estimate effects each month during the sample period to better understand the dynamics of the response. We run the following regression:

$$\pi_{i,t} = \alpha_i + \delta_t + \beta_t^{\text{Direct}} s_i^{\text{Direct}} + \beta_t^{\text{Indirect}} s_i^{\text{Indirect}} + \gamma \hat{w}_{i,t-1} + \varepsilon_{i,t}, \tag{12}$$

where predicted shocks are now calculated using only the constant-percentage markup assumption, and where we include expenditure categories for both goods and services. Additionally, this regression separates predicted effects from direct and indirect tariff exposure to better compare the two and understand the timing of each.

Results are shown in Figure 4; panel (a) shows the coefficient on the predicted direct effect over time, and panel (b) shows the coefficient on the predicted indirect effect. We see significant results from the direct effect starting in September 2018, with prices continuing to rise for affected expenditure categories through the beginning of 2019. The coefficient hovers around 1, which suggests full pass-through under the constant-percentage markup assumption.We also see that, for the most part, the parallel-trends assumption seems to hold, though there may have been anticipation effects in April 2018.[17] The indirect effect appears later, with significant results starting in December 2018. The coefficient on the indirect effect is large, but so are standard error bars. This makes sense, given that we would expect more noise as tariffs work their way through the supply chain.

Figure 4: Estimated Pass-through of 2018 China Tariffs



(a) Direct

(b) Indirect

Notes: This figure examines the pass-through from the China tariffs enacted during the July–September 2018 period. Panel (a) shows the coefficient on the predicted direct effect, and panel (b) shows the coefficient on the predicted indirect effect. Standard errors are clustered at the NIPA expenditure category level. The dotted red line shows the last month before the post-period. The blue bars represent the 90 percent confidence intervals. *Sources*: BEA, BLS, Fajgelbaum et al. (2019), authors' calculations.

4 Forecasting Inflation with the Consumption-weighted Import Price Index

How different is our methodology from a simple import-weighted tariff or import price index in predicting consumption price pass-through? The previous section showed how using

[17]The uptick in 2018 may also have been caused by contamination by earlier tariffs, namely the washing machine tariffs enacted in February 2018 and the steel and aluminum tariffs enacted during the March–June 2018 period.

cost-based shares—which is our full pass-through assumption—improves the fit of a tariff shock pass-through. In this section, we show that our methodology improves the prediction of the import-price pass-through into PCE more generally by comparing it with the official BLS Import Price Index. Unlike our index, in which weights are based on the import price sensitivity of consumption, the BLS Import Price Index weights goods with trade dollar value shares. However, not only are goods imported for final consumption or as an intermediate for final consumption production, but also imports are used for government expenditure, investment, exports, and more. In other words, input–output table accounting is necessary to isolate both cost-based exposures and the effects on the growth component of interest.

Constructing the Consumption-weighted Import Price Index We exploit our methodology to build an alternative import price index, tailored to estimate the effect of import prices on final consumption. To do so, we build a concordance matrix C between the most disaggregated industry import codes available from the BLS Import Price Index and BEA commodity codes. We start with the BEA-NAICS concordance to group various BEA codes into a single NAICS code. We use NAICS codes that correspond to the most disaggregated NAICS level of the BLS import price series.[18] If a BEA commodity does not map to a BLS series, we drop that commodity from this analysis, which translates to an assumption of zero import price growth.

Given this concordance matrix as C, the contribution of imports to PCE inflation can be defined as

$$\hat{p}_t^{\text{Import}} = \mathbf{1}^\top(\Delta_{t-12}^{\text{Direct}} + \Delta_{t-12}^{\text{Indirect}})C\hat{p}_t^*,$$

where p_t^* comes from the import price indices published by the BLS and sensitivities are constructed using the constant-percent markup assumption. To turn this into an index, we reinterpret our sensitivities as shares and simply rescale these shares so they sum to one within a date across NAICS codes. Thus, our PCE import price index is defined as

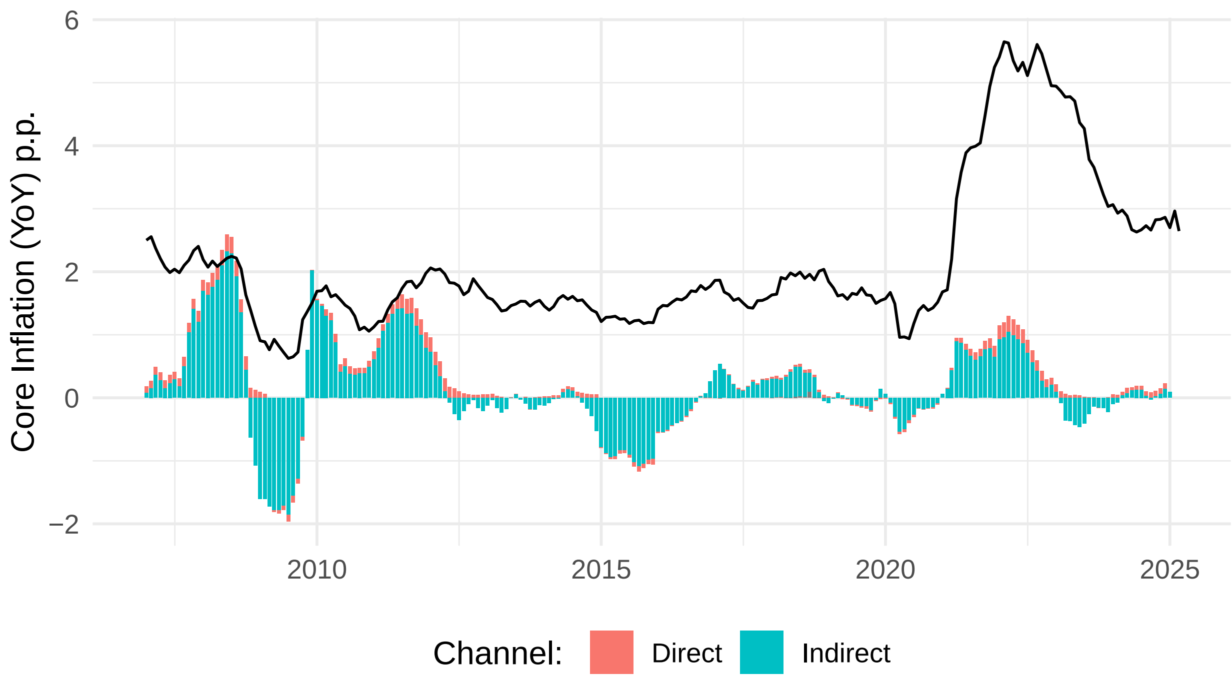
$$\text{IPI}_t^{\text{PCE}} = \mathbf{1}^\top W_{t-12}^c(\Delta_{t-12}^{\text{Direct}} + \Delta_{t-12}^{\text{Indirect}})C\hat{p}_t^*,$$

where $\hat{p}_t^*$ is a vector containing disaggregated values of the import price index from the BLS for month t. The sensitivity shares date to the previous year's share to map closely to how the aggregate price import is computed by the BLS.

[18]For example, the BEA code 1111A0 includes NAICS codes 11111 and 11112, so we map this BEA code to the NAICS code 1111. However, the BLS does not include an import series for NAICS 1111, only 111. Thus, BEA code 1111A0 corresponds to BLS code 111.

The Contribution of Import Price Changes to Inflation Figure 5 shows the contribution of import prices to core PCE inflation under the full pass-through assumption from 2007 through 2024. This contribution is broken down into the direct and indirect import channels, depicted by the red and blue bars, respectively. For reference, core PCE inflation is shown as the black line. Notably, the higher indirect contribution component is not due to the fact that indirect effects of import prices account for a higher share of PCE than direct effects, but rather that categories with a high indirect contribution have more *volatile* import prices.

Figure 5: The Contribution of Import Price Changes to Core PCE Inflation



Notes: The import price contribution is computed from disaggregated import prices weighted according to our import sensitivities. The red bars show price increases due to direct imports, and the blue bars show those due to indirect imports. The black line shows core PCE inflation. *Sources*: BEA, BLS, authors' calculations.

Assessing the Performance of the Consumption-weighted Import Price Index Given that we use exactly the same underlying time series $\hat{p}_{m,t}$ as the BLS to compute our import PCE index, our methodology differs in only the relative weighting given to different sectors. Why do we expect better performance from our index? Unlike our index, in which weights are based on the import price sensitivity of consumption, the BLS Import Price Index weights goods with trade dollar value shares. However, as noted, not only are goods imported for final consumption or as an intermediate for final consumption production, but also imports are used for government expenditure, investment, exports, and more.

We thus turn to assessing the forecasting ability of our consumption-weighted import price index. Specifically, we compare its forecasting ability to the BLS Import Price Index, excluding fuel and food. Unlike the BLS Import Price Index, our index takes food and energy into account in predicting core PCE as indirect imports. In other words, food and energy price changes may impact our overall index through the supply chain. The "core" BLS import price does not include any impact of food or energy.

We implement h-month-ahead forecasts of core PCE with both our index and the "core" BLS Import Price Index, using information starting in 2007:m1 and reestimating the model every month from 2017:m1 through 2024:m12:

$$p_{t+h} - p_{t+h-1} = \alpha + \sum_{j=0}^{11} \beta_j (z_t - z_{t-1}) + \sum_{j=0}^{11} \mu_j (p_{t-j} - p_{t-j-1}) + \varepsilon_{t+h}, \tag{13}$$

where p denotes the core PCE price index in a given month and z denotes the relevant index. For various horizons of $h = 1, 2, 3, 6, 12$ months, we report the RMSE ratio between our baseline model, in which z denotes our consumption-weighted import price index, and the competing model, in which z denotes the "core" BLS Import Price Index. We also run a Diebold–Mariano test between these two competing models and report the test statistic and p-value. We then report the additional R^2 that our model delivers over the entire period via an alternative model:

$$p_{t+h} - p_{t+h-1} = \alpha + \sum_{j=0}^{11} \mu_j (p_{t-j} - p_{t-j-1}) + \varepsilon_{t+h}. \tag{14}$$

Results are reported in Table 2. We see that at all five horizons, our baseline model outperforms the competing model in which our consumption-weighted import price index is replaced with the core BLS Import Price Index.[19] For horizons of one, two, and three months, this outperformance is statistically significant.

[19] Our index constructed with the constant-percentage markup assumption also outperforms the core BLS Import Price Index. This supports the importance of isolating the consumption contribution of the shares, even when not using cost-based shares, which map to the constant-percentage markup assumption.

Table 2: The Predictive Power of Import Prices on Core PCE Inflation

			Diebold–Mariano Test	
Horizon	Additional R2	RMSE Ratio	Statistic	p-value
1	0.0315	0.909	-1.738	0.0822
2	0.0542	0.879	-2.448	0.0144
3	0.0512	0.876	-2.254	0.0242
6	0.0477	0.844	-1.244	0.2134
12	0.0657	0.588	-1.334	0.1823

Notes: Additional R2 is the difference between the R2 of the model using our index and the R2 of the model without an additional predictor z. RMSE is the square root of the average prediction error; the RMSE ratio is the ratio of the RMSE of the model with our index versus the RMSE of the model with the "core" BLS Import Price Index. The null hypothesis of the Diebold–Mariano test holds that a variable has the same forecast accuracy of the "core" BLS Import Price Index. A negative statistic indicates that the model with our index is a better predictor. *Sources*: BEA, BLS.

5 Conclusion

In this paper, we study the effects of import prices at the border on US consumer prices. We do so by developing a new methodology that builds on a static open economy model of firms using domestic and foreign goods in a network structure. To that model we add markups and retailers. We carefully map each model object to official, publicly available US data, thus constructing matrices that show the sensitivity of each consumer expenditure category to import price changes in underlying commodities.

We show how these matrices have utility for this, and future, research. First, we can use the matrices to understand the share of imported value added in PCE, decomposed into expenditure categories and into direct and indirect imports. Second, we can assess the partial-equilibrium effect of various tariff scenarios, a crucial tool for policymakers and business leaders as tariff schedules are updated. Third, we show that our network mapping between import prices and retail prices reveals full pass-through of the 2018 tariffs into consumer prices. Finally, we can calculate a consumption-weighted import price index, which, as we show outperforms, the "core" BLS Import Price Index in inflation forecasting.

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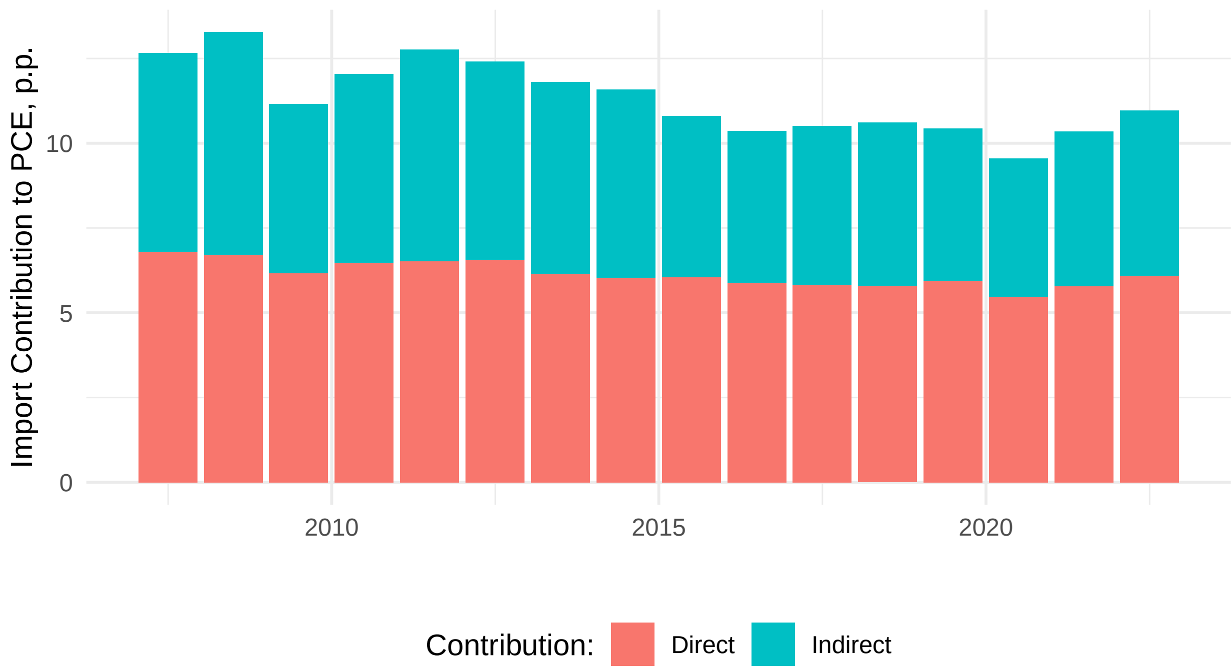
A Additional Figures

Figure A.1: PCE Shares of Imports over Time



Notes: The red bars show the share of PCE corresponding to imported goods that are directly consumed by US households. The blue bars show the share of PCE corresponding to imports that are used as inputs in domestic production. *Sources*: US Bureau of Economic Analysis and authors' calculations.